

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 5

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Subject Name : Deep Learning

Problem Statement

Implement and compare Simple RNN, LSTM, and GRU models for sequence classification using the IMDb movie review dataset, and analyze their performance using appropriate evaluation metrics.

Algorithm

1. Begin.
2. Load the IMDb movie review dataset for binary sentiment classification.
3. Set the vocabulary size to 10,000 words.
4. Set the maximum sequence length to 200 tokens.
5. Pad and truncate the review sequences to the fixed length.
6. Visualize the distribution of review sequence lengths.
7. Initialize a Simple RNN model with an Embedding layer and 64 RNN units.
8. Apply Dropout with a rate of 0.5.
9. Add a Sigmoid output layer for binary classification.
10. Compile the RNN using the Adam optimizer and binary cross-entropy loss.
11. Train the RNN model for 5 epochs and record the training time.
12. Evaluate the RNN using accuracy, precision, recall, F1-score, and confusion matrix.
13. Initialize an LSTM model with an Embedding layer and 64 LSTM units.
14. Apply Dropout and a Sigmoid output layer.
15. Compile and train the LSTM model for 5 epochs.
16. Evaluate the LSTM using accuracy, precision, recall, F1-score, and confusion matrix.
17. Initialize a GRU model with an Embedding layer and 64 GRU units.
18. Apply Dropout and a Sigmoid output layer.
19. Compile and train the GRU model for 5 epochs.
20. Evaluate the GRU using accuracy, precision, recall, F1-score, and confusion matrix.
21. Plot training and validation accuracy and loss for all three models.
22. Compare RNN, LSTM, and GRU using accuracy, precision, recall, F1-score, and training time.
23. Display the comparative performance using graphs.
24. Determine the relative performance of the three sequence classification models.
25. End.

OUTPUT:

RNN vs LSTM vs GRU — Sequence Classification

